



VVCE

VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute | Accredited by NAAC with 'A' Grade

Gokulam 3rd Stage, Mysuru - 570002

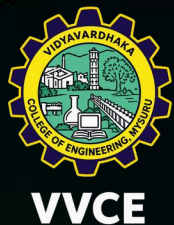


IN7.0 THON

« Architecting an Autonomous Tomorrow. »

✦ TEAM NAME:EcoLogic

DOMAIN :Agentic AI ✦



PROBLEM IDENTIFICATION

1,77,175

Road fatalities in India (2024) — 1 life lost every 3 minutes

Source: MoRTH, Road Accidents in India 2024

2,385 ▲53%

Pothole deaths in 2024 alone — 9,438 in 5 years

Source: MoRTH / Supreme Court Committee on Road Safety

#2 13.9 km/h

Bengaluru: most congested city on Earth · 36 min per 10 km

Source: TomTom Traffic Index 2025

THE PROBLEM: Authorities govern multi-thousand-km road networks **blind** — reacting to complaints and manual patchwork while hazards claim lives and transit crawls.

4 ROOT CAUSES

Reactive Patchwork

01

Repairs happen only after accidents or complaints; decay outpaces maintenance.

96% Blind Spots

02

Fixed AI cameras cover <4% of corridors; the rest is unmonitored.

Agency Data Silos

03

Traffic police and PWD run disconnected systems with no shared map.

60-Min Triage Lag

04

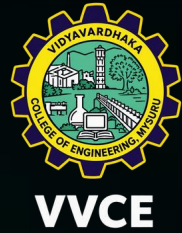
Floods, debris and stalls reach dispatch after an hour; gridlock grows.

GAP ANALYSIS — WHY EXISTING SOLUTIONS FALL SHORT

EXISTING SOLUTION	WHAT IT DETECTS	CRITICAL GAP VS. URBANSIGHT AI
Hayden AI (US/EU)	Bus-lane & parking violations	No hazard intelligence — blind to potholes & flooding
Rebotnix (Germany)	Static road distress, signs	No live incidents — tied to garbage routes
BBMP Survey Vans	One-off road damage survey	No continuity — data goes stale in weeks
Indian Startups	Potholes via phone vibration	Siloed — no command integration or triage
Karnataka AI Cameras	Speeding, seatbelt, lane	Fixed-point blindness — <4% coverage, ₹Cr/km

WHY NOW

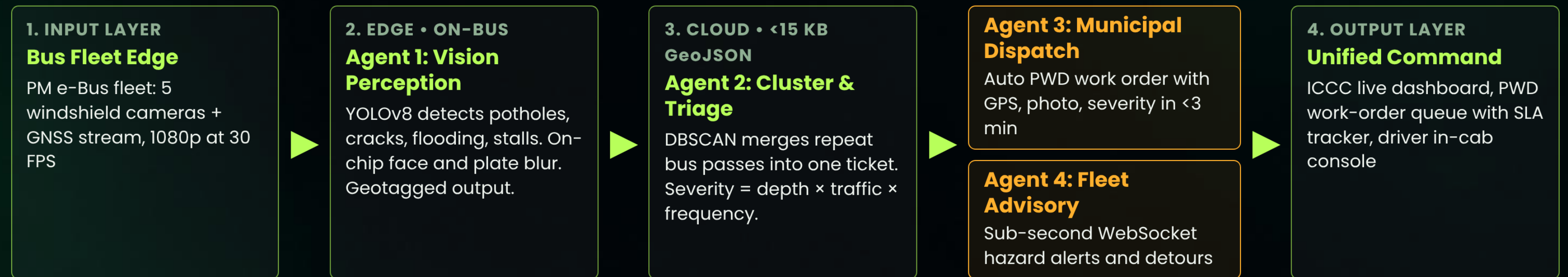
10,000 PM e-Buses with 5 CCTV cameras, GPS & 4G roll out across 169 cities — **a ready mobile sensing grid with ZERO new sensor CapEx.**



PROPOSED SOLUTION

UrbanSight AI turns existing city bus dashcams into continuous mobile scanning grids: on-edge vision detects hazards, and autonomous agents close the loop from **detection to PWD repair dispatch in under 3 minutes**.

4-Agent Pipeline



Root Cause → Resolution

Root Cause	UrbanSight Solution	Impact
Periodic, reactive monitoring	Continuous daily edge scanning by buses	20-50+ inspections/day
Data siloed by hazard type	Unified multi-class vision pipeline	1 pane of glass for 3 systems
Static camera blind spots	Mobile transit mesh across arterials	85%+ arterial reach , no poles
Detection never closes the loop	Agent 3 autonomous dispatch	<3 min to official PWD ticket

Innovation Edges

Tri-Vector Perception: road damage, traffic stagnation and live incidents in one compute loop.

Privacy by Design: faces and plates blurred on-chip; only anonymized GeoJSON leaves the bus.

Zero Hardware CapEx: uses cameras and GPS already mandated on 10,000+ PM e-Buses.



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TECHNOLOGY STACK

6-LAYER FUNCTIONAL STACK

L1
LAYER

Edge Sensing & Compute

Jetson Orin Nano · 40 TOPS RPi 4 + Coral TPU · frugal 1080p 30 FPS Dashcam

Rugged 7–15W compute, retrofit under ₹15,000/bus.

L2
LAYER

Edge AI & Privacy Engine

YOLOv8n / YOLO26s · TensorRT MobileNetV2 OpenCV face & ANPR blur

Sub-30ms inference at 30 FPS; anonymized on-device first.

L3
LAYER

Telemetry & Transport

MQTT over TLS 1.3 Mosquitto Broker Pydantic v2 · GeoJSON

Encrypted <15 KB JSON per event — zero video bandwidth.

L4
LAYER

Cloud Ingestion & Backend

FastAPI (ASGI) Python 3.13 Docker · Railway / GCP Run

Async REST + WebSocket gateway, scales to zero (₹800–1,500/mo).

L5
LAYER

Geospatial Intelligence

DBSCAN · scikit-learn PostGIS · PostgreSQL 16 Uber H3 · Res 9

Merges repeat detections into one confidence-weighted ticket.

L6
LAYER

Command Center & APIs

React 18 + Leaflet.js CartoDB Dark WebSockets BBMP / PWD REST

Sub-second live map; evidence-backed auto-complaints.

WHY THIS CHOICE?

◆ Edge Compute + Privacy (YOLOv8n + OpenCV)

99.9% less bandwidth: 100 buses = ~5 TB/day raw video vs <15 KB GeoJSON per detection. **DPDP-compliant:** faces & plates blurred in edge RAM — zero PII in cloud.

◆ Spatial Dedup (DBSCAN + H3)

30 buses, same pothole → **one** PWD work order, not 30. Groups detections within 5 m and raises severity confidence.

◆ Asymmetric Hardware (Jetson vs Coral)

Survey vans cost ₹40L–₹1Cr+. Jetson on busy routes, RPi + Coral on feeders → **>80% CapEx cut.**

📊 DATASET STRATEGY

Gap: no public bus-windshield dataset. **Base:** RDD2022 (47,420 images, 55,000+ annotations). **Fix:** BDD100K transfer + windshield pitch/yaw & monsoon/shadow augmentation.

100%
open-source

₹800–1.5K
/mo · 10 buses

₹8–15K
/bus · 1/50th of a van



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FEASIBILITY & IMPACT

75% FASTER

PWD Repair Dispatch

Baseline: 14-21 days → UrbanSight: **<48 hours**

40% REDUCTION

Road-Hazard Fatality Risk

Baseline: 2,385 deaths (2024) → **pre-failure repair**

85%+ COVERAGE

City Arterial Network

Baseline: <15% from 7 fixed cameras → **mobile mesh**

Phased Roadmap

Phase	Milestone	Proof Gate
1. Prototype (Now) 1-2 buses	YOLOv8n on Jetson, DBSCAN, live Leaflet dashboard	>82% mAP50 on RDD2022 India; <30 ms edge latency
2. City Pilot (3-6 mo) 10-50 buses	Auto-complaint bridge to PWD/BBMP portal, driver console	500+ verified detections; 92% precision vs PWD audits
3. National (6-18 mo) 500 to 10,000 buses	City heatmaps, predictive deterioration model, state ITMS API	Transit MoUs; 60% lower long-term maintenance cost

Risks & Mitigations

Risk	Mitigation
Privacy / DPDP	On-chip face and plate blur; only anonymized <15 KB GeoJSON sent
Route coverage bias	Arterials first (80%+ of fatalities); sanitation fleet and citizen app fill gaps
Harsh Indian conditions	3-frame consensus plus IMU vibration cross-check; RDD2022 fine-tuning
Moving-bus data scarcity	BDD100K transfer learning, windshield augmentation, human review loop
PWD adoption	Headless API injects tickets into existing portals; no new software

B2G SaaS & Market Size

₹80,000/month per transit authority

169 cities × ₹80K × 12 = ₹16.22 Cr/yr SAM

10 tier-1 cities in Year 1 = **₹96 L ARR**; 88%+ gross margin

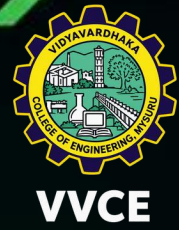
UN SDG Alignment

SDG 11.2: 40% fewer hazard casualties on monitored corridors

SDG 11.3: 85%+ arterial coverage; dispatch in <48 h

SDG 9.1: 60% lower lifecycle maintenance cost per lane-km

UrbanSight AI converts the **10,000 e-buses** already rolling across India into an autonomous national road intelligence grid: **detecting hazards, ranking risks, and filing repair orders** before the next pothole takes another life.



RESEARCH & REFERENCES

RESEARCH & DATA SOURCES

- MoRTH — Road Accidents in India 2024
- TomTom Traffic Index 2025
- RDD2022 — Road Damage Dataset
- BDD100K — Driving Dataset
- PM e-Bus Sewa — 10,000 e-bus scheme
- DPDP Act 2023 — Data privacy
- YOLOv8 · scikit-learn DBSCAN · Uber H3

PROJECT LINKS — CLICK TO OPEN

PROJECT REPORT

Google Drive folder

DEMO VIDEO

Google Drive folder

GITHUB REPOSITORY

github.com/gaganks2006-ind/urbansight-ai

LIVE FRONTEND

<https://gaganks2006-ind.github.io/urbansight-ai/>

